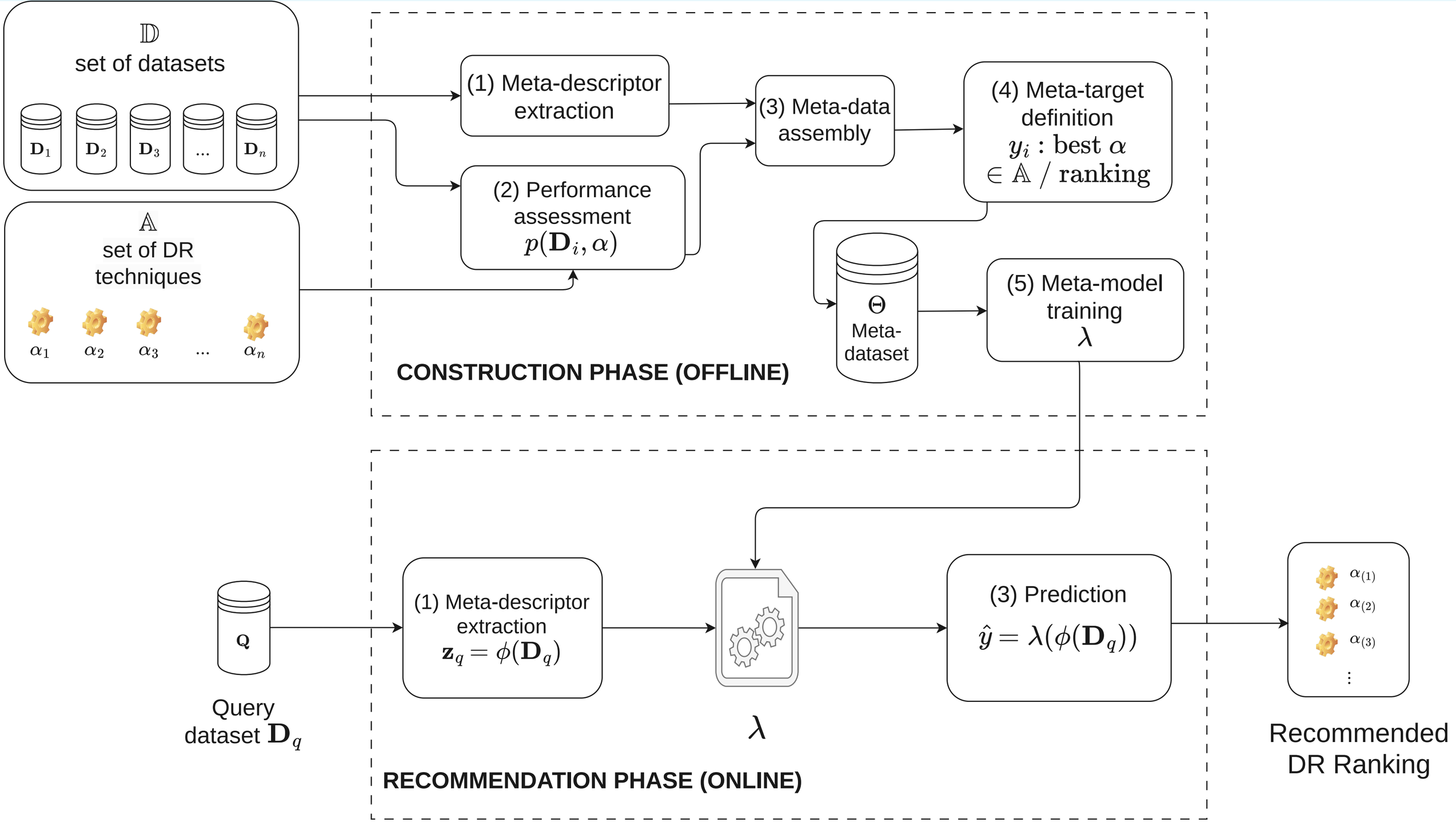
$\mathbb{D}$
set of datasets
$\mathbf{D}_1$ $\mathbf{D}_2$ $\mathbf{D}_3$ ... $\mathbf{D}_n$
$\mathbb{A}$
set of DR techniques
α_1 α_2 α_3 ... α_n
(1) Meta-descriptor extraction
(2) Performance assessment $p(\mathbf{D}_i, \alpha)$
(3) Meta-data assembly
(4) Meta-target definition y_i : best α $\in \mathbb{A}$ / ranking
Θ Meta-dataset
(5) Meta-model training λ
CONSTRUCTION PHASE (OFFLINE)
Q
Query dataset $\mathbf{D}_q$
(1) Meta-descriptor extraction $\mathbf{z}_q = \phi(\mathbf{D}_q)$
λ
(3) Prediction $\hat{y} = \lambda(\phi(\mathbf{D}_q))$
$\alpha_{(1)}$ $\alpha_{(2)}$ $\alpha_{(3)}$ ⋮
Recommended DR Ranking
RECOMMENDATION PHASE (ONLINE)